Lecture 13: Related Rates I

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Exam 2 Soon!

Exam 2 is coming up! It is two Tuesday's from now, 11/3.
It's covering linearizations + our derivative rules + related rates + L'Hôpital's Rule which we'll cover next week.

Quiz Time

Let's get into neat rows, spread out, and slay this quiz!

Implicit Differentiation Recap

Last lecture we covered *implicit differentiation*, the process of relating variables using some implicit relation and differentiating that relation to get a relation between their derivatives.
E.g.: slope of line tangent to $y^2 = x^3 - 10x + 1$ at $(0, 1)$.

Slope of tangent line $\implies$ interested in $\dfrac{dy}{dx}$ $\implies$ treat y as a function of x.

$$y^2 = x^3 - 10x + 1 \tag{1}$$

$$\frac{d}{dx}y^2 = \frac{d}{dx}(x^3 - 10x + 1) \tag{2}$$

$$\frac{dy}{dx}\frac{d}{dy}y^2 = 3x^2 - 10 \tag{3}$$

$$2yy' = 3x^2 - 10 \tag{4}$$

$$2 \cdot 1 \cdot y' = 3 \cdot 0^2 - 10 \tag{5}$$

$$2y' = -10 \quad y' = -5 \tag{6}$$

Towards Related Rates

Suppose instead that some particle was moving over time in the $2D$ plane and whose motion was constrained to that curve cut out by $y^2 = x^3 - 10x + 1$.

- Moving over time $\implies$ it's position can be described by functions of a time variable t
- In the $2D$ plane $\implies$ it has an x-position $x(t)$ and a y-position $y(t)$
- Motion constrained $\implies$ $y(t)^2 = x(t)^3 - 10x(t) + 1$.

In order to stay on that curve, what restrictions does that place on the particle's:

1. Position?
2. Velocity?
3. Acceleration?

Relating Some Rates I

Let's see what our constraint on position forces upon our velocity!

$$y(t)^2 = x(t)^3 - 10x(t) + 1 \tag{7}$$

Differentiate w.r.t. t to get velocity (8)

$$\frac{d}{dt}y(t)^2 = \frac{d}{dt}(x(t)^3 - 10x(t) + 1) \tag{9}$$

$$\frac{dy}{dt}\frac{d}{dy}y^2 = \frac{dx}{dt}\frac{d}{dx}(x^3 - 10x + 1) \tag{10}$$

$$2y\frac{dy}{dt} = (3x^2 - 10)\frac{dx}{dt} \tag{11}$$

Relating Some Rates II

So if the particle is at $(0, 1)$ at some time $t = t_0$ and has a horizontal velocity of $3m/s$ which means $(x(t_0), y(t_0)) = (0, 1)$ and $\frac{dx}{dt}\,|_{t=t_0} = 3$ what does that force the vertical velocity to be?

$$2y\frac{dy}{dt} = (3x^2 - 10)\frac{dx}{dt} \tag{12}$$

$$2 \cdot 1\frac{dy}{dt} = (3 \cdot 0^2 - 10) \cdot 3 \tag{13}$$

$$2\frac{dy}{dt} = -30 \quad \frac{dy}{dt} = -15m/s \tag{14}$$

(Notice from before, we found the slope of the line tangent to the curve at $(1, 0)$ was -5 and $-15 = -5 \cdot 3$, so we're exactly enforcing that the velocity is parallel to the tangent line).

Related Rates I

This process is common enough that we call problems using it *related rates* problems:

Related Rates

Given a real-world/physical problem that involves:

- Some number quantities and measurements for them
- Some number of implicit relations between them
- Some information about the rate of changes of some of the quantities

and seeks information about the rate of changes of the other variables
solving the problem via the method of related rates involves:

1. Differentiating the relations between the quantities with respect to the variable of interest
2. Substituting in the known quantities
3. Solving the resulting equation(s) for the rate of change in question.

Related Rates Example I

For a classic related rates problem: Suppose you have a cylindrical tank of water whose height is 10 meters and whose radius is 2 meters, with a hole at the bottom from which water drains out. At the moment the height of the water is halfway to the bottom, a hole at the bottom of the tank opens and you measure the speed at which the water escapes to be about $\frac{1}{2}\frac{m^3}{sec}$. At what rate will the height of the water decrease?
Identify information:

- Quantities: volume, height, radius. Radius is constant at 2 meters, volume changes w.r.t. height. Height is at 5 meters.
- Relations: $V = \pi r^2 h$.
- RoC information: $\frac{dV}{dt} = -\frac{1}{2}\frac{m^3}{sec}$.
- RoC in question: $\frac{dh}{dt}$.

Related Rates Example I cont.

We then want to differentiate the relation (volume in terms of radius and height) with respect to time to relate the rates of change:

$$V = \pi r^2 h \tag{15}$$

$$\frac{d}{dt}V = \frac{d}{dt}(\pi r^2 h) \tag{16}$$

$$\frac{dV}{dt}\frac{d}{dV}V = \pi \frac{d}{dt}(r^2 h) \tag{17}$$

$$\frac{dV}{dt} \cdot 1 = \pi \cdot (\frac{d}{dt}(r^2) \cdot h + r^2 \cdot \frac{d}{dt}(h)) \tag{18}$$

$$\frac{dV}{dt} = \pi \cdot (\frac{dr}{dt} \cdot (\frac{d}{dr}r^2) \cdot h + r^2 \cdot \frac{dh}{dt}\frac{d}{dh}h) \tag{19}$$

$$\frac{dV}{dt} = \pi \cdot (0 \cdot _ + r^2 \frac{dh}{dt}) \quad \Longrightarrow \quad \frac{dV}{dt} = \pi r^2 \frac{dh}{dt} \tag{20}$$

$$-\frac{1}{2} = \pi \cdot 2^2 \frac{dh}{dt} \quad \Longrightarrow \quad \frac{dh}{dt} = -\frac{1}{8\pi}\frac{m}{sec} \tag{21}$$

So the height of the water is decreasing at a rate of $\frac{1}{8\pi}$ meters per second.

Related Rates Example II

Now suppose that the tank is instead an *inverted cone* with the same height and radius. What changes?

- The volume formula changes: $V = \frac{1}{3}\pi r^2 h$
- AND the radius also changes over time:

Giving us the *relation*:

$$V = \frac{1}{3}\pi r^2 h \tag{22}$$

$$V = \frac{1}{3}\pi \left(\frac{2}{10}h\right)^2 \cdot h \tag{23}$$

$$V = \frac{4\pi h^3}{300} = \frac{\pi h^3}{75} \tag{24}$$

Related Rates Example II cont.

We would then differentiate this relation w.r.t. t to get:

$$\frac{d}{dt}V = \frac{d}{dt}\frac{\pi h^3}{75} \tag{25}$$

$$\frac{dV}{dt} = \frac{\pi h^2}{25}\frac{dh}{dt} \tag{26}$$

$$\frac{dh}{dt} = \frac{25}{\pi h^2}\frac{dV}{dt} = \frac{25}{\pi \cdot 5^2} \cdot \frac{-1}{2} = \frac{-1}{2\pi}\frac{m}{s} \tag{27}$$

So now the height is decreasing at a rate of $\frac{1}{2\pi}$ meters per second.

Related Rates Example III

For a non-tank draining problem, let's instead suppose there is a 10 meter long ladder propped up at an angle on a wall that makes a right angle with the ground. It's base is currently 6 meters away from the wall, and that base is sliding away from the wall at a rate of two meters per second. At that time, how high up along the wall is the ladder, and what is the rate of change of that height?

- Quantities: It is very helpful here to draw a picture!
  Our quantities are then the distance of the base of the ladder away from the wall $x(t)$ and how high along the wall the ladder is $y(t)$.
- Relations: The length of the ladder is constant $\implies x(t)^2 + y(t)^2 = 10^2$.
- RoC information: $\dfrac{dx}{dt} = 2\dfrac{m}{s}$.
- Information in question: value of $y(t)$ and $\dfrac{dy}{dt}$.

Related Rates Example III cont.

We differentiate the relation w.r.t time:

$$x^2 + y^2 = 10^2 \tag{28}$$

$$\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}10^2 \tag{29}$$

$$\frac{d}{dt}(x^2) + \frac{d}{dt}(y^2) = 0 \tag{30}$$

$$\frac{dx}{dt}\frac{d}{dx}x^2 + \frac{dy}{dt}\frac{d}{dy}y^2 = 0 \tag{31}$$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \tag{32}$$

$$2 \cdot 6 \cdot 2 + 2 \cdot y \cdot \frac{dy}{dt} = 0 \tag{33}$$

$$\frac{dy}{dt} = -\frac{12}{y}\frac{m}{sec} \tag{34}$$

Related Rates Example III cont. 2

$$\frac{dy}{dt} = -\frac{12}{y}\frac{m}{sec} \tag{35}$$

So we still need to determine what $y(t)$ is when $x(t) = 6$. How?
Use the relation!

$$x(t)^2 + y(t)^2 = 10^2 \tag{36}$$

$$6^2 + y(t)^2 = 100 \tag{37}$$

$$y(t)^2 = 100 - 36 \tag{38}$$

$$y(t)^2 = 64 \quad \Longrightarrow \quad y(t) = \pm 8 \tag{39}$$

and distances must be positive so $y(t) = 8$ when $x(t) = 6$.
So:

$$\frac{dy}{dt} = -\frac{12}{y} = -\frac{12}{8} = -\frac{3}{2}\frac{m}{sec}. \tag{40}$$

The height of the ladder against the wall is therefore decreasing at a rate of $\frac{3}{2}$ meters per second.

More to Come!

We'll also be spending next Tuesday going over more examples of how to use the method of related rates to solve different types of problems.
Generally however, all of these problems will rely on some kind of geometric input, e.g.:

- Area of square, rectangle, circle, ellipse, triangle
- Perimeter of the above (except ellipse)
- Angles (e.g. SOHCAHTOA trig)
- Volumes of cubes, rectangular prisms, pyramids, cones, spheres
- Surface areas of the cubes, rectangular prisms, spheres
- Pythagorean theorem
- Similar triangles

and thus having some way to commit these formulae to memory will be useful.

Worksheet Time

Let's get into groups and work on the worksheet!

End of Class

That’s all folks!